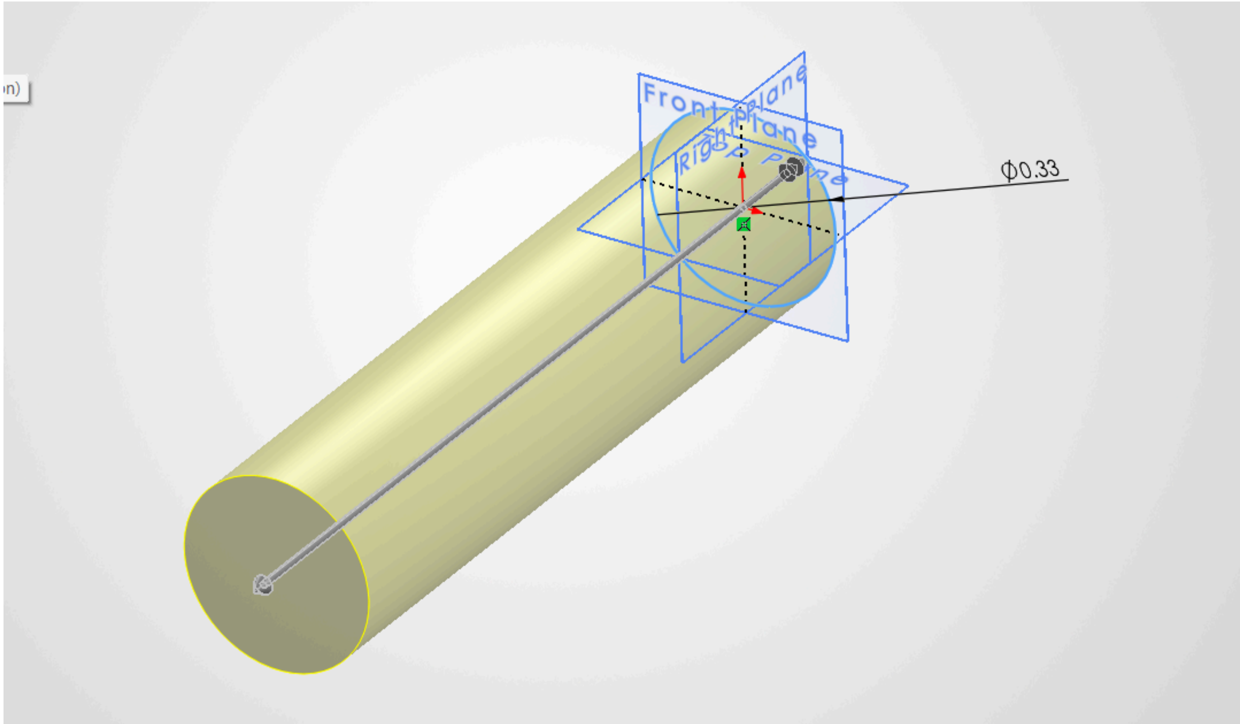
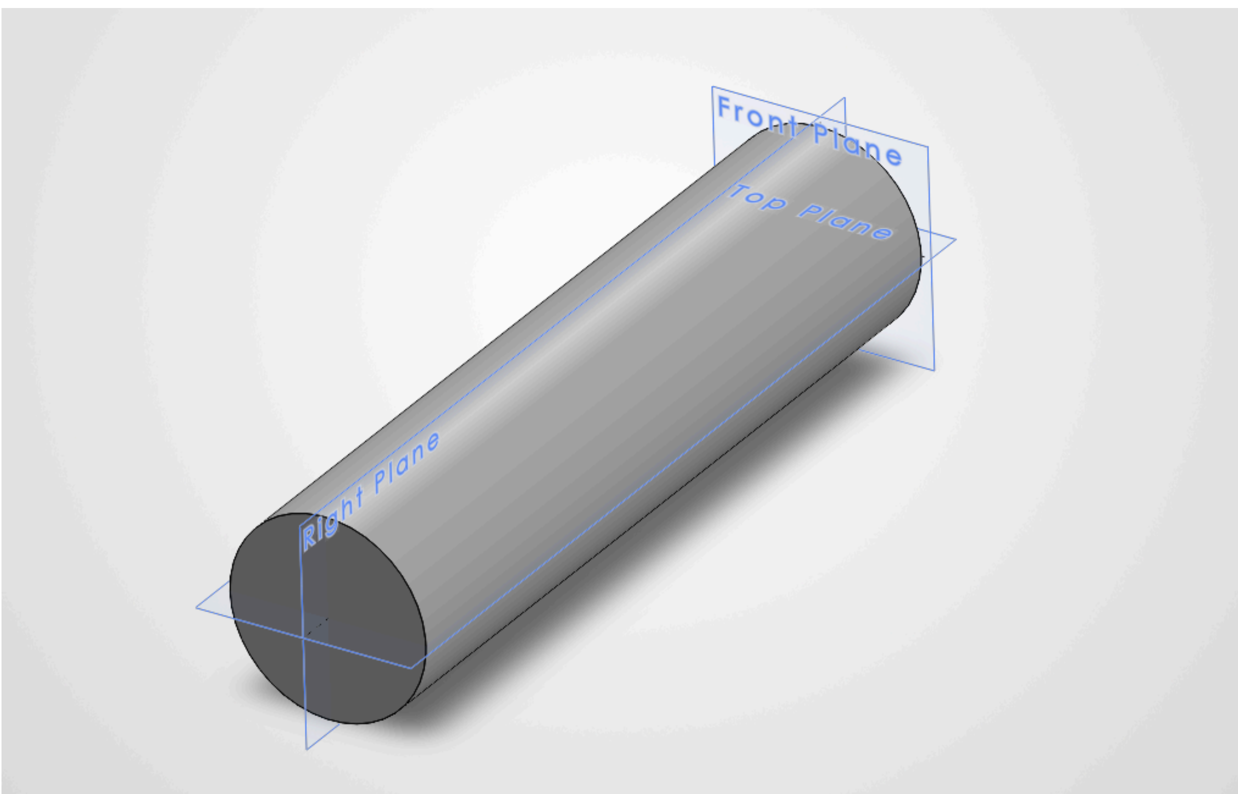
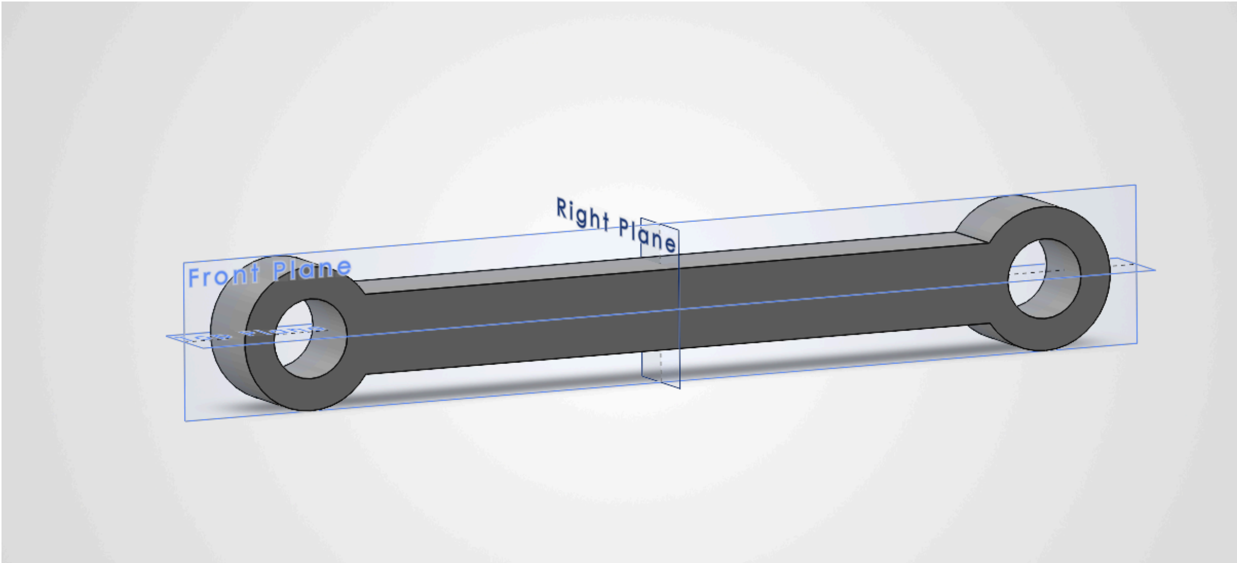
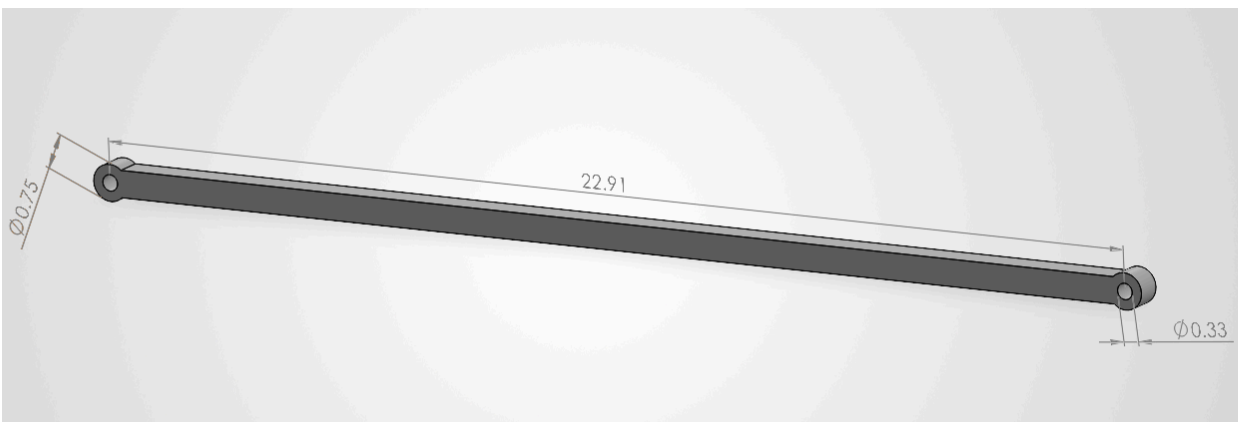


Pin 1: Single truss connection, pin length 1

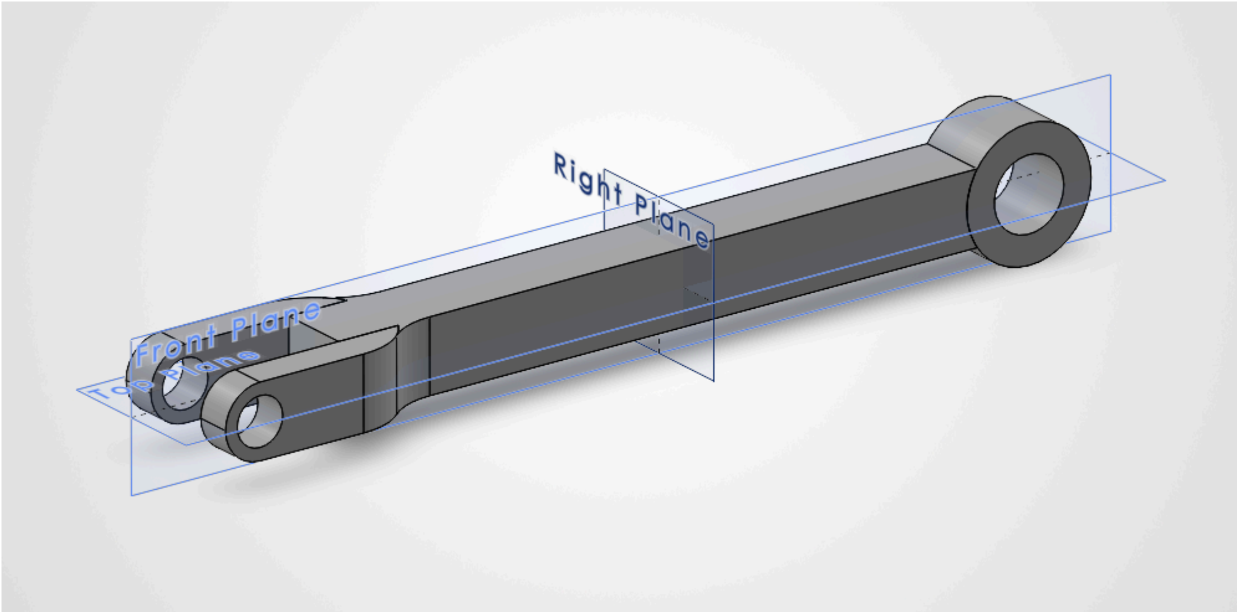




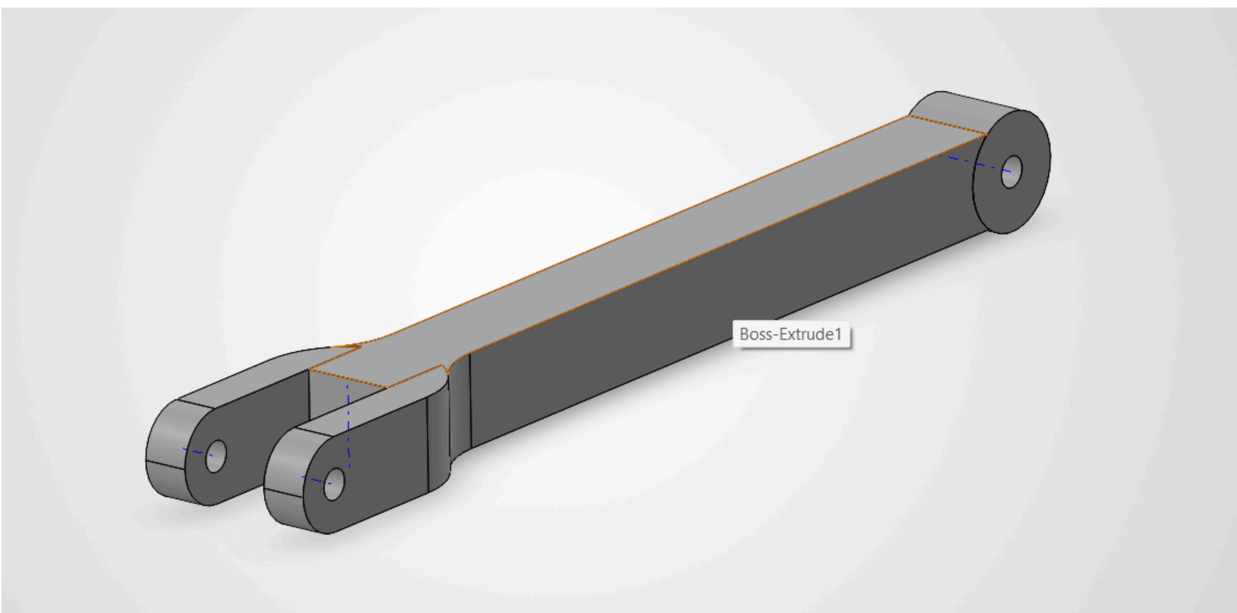
Single truss member (prototype)



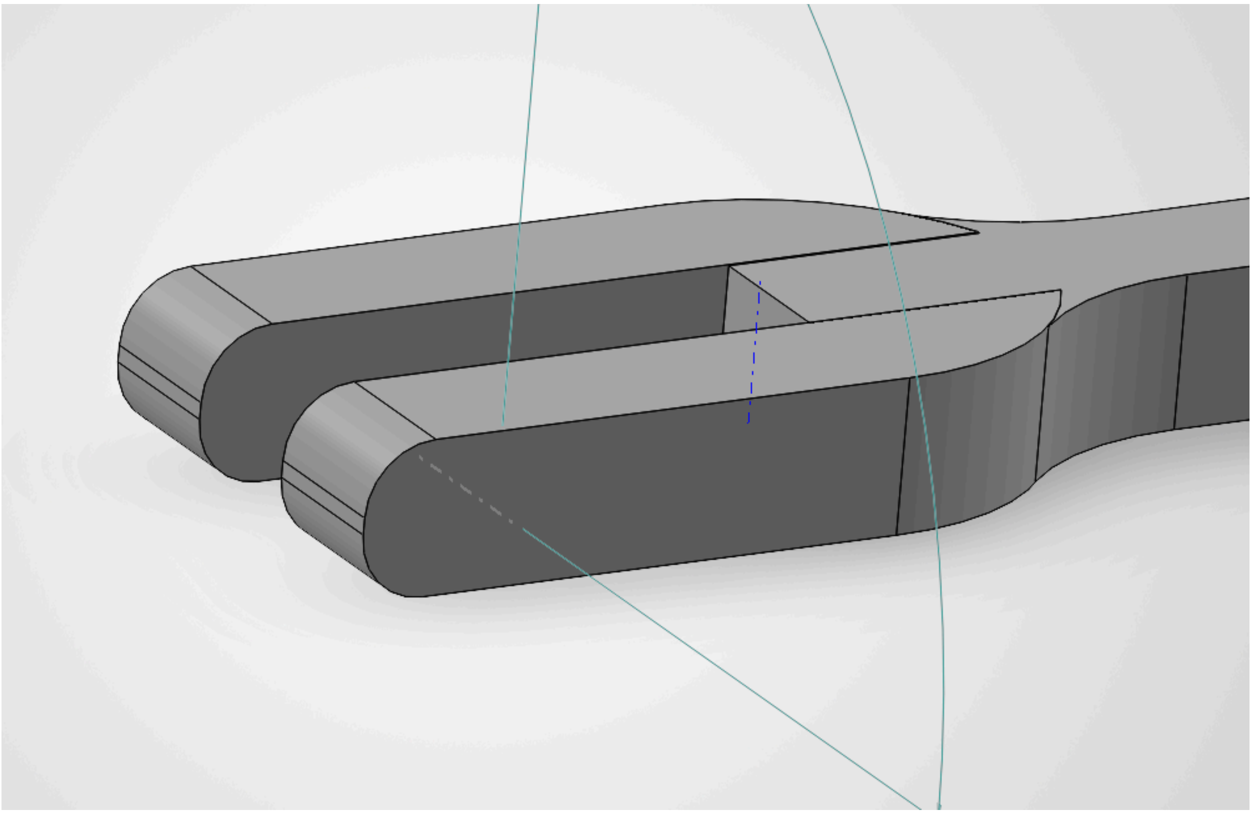
Redimensioning single truss member to fit cross sectional area, length and pin diameter (0.6m = 23.62in)



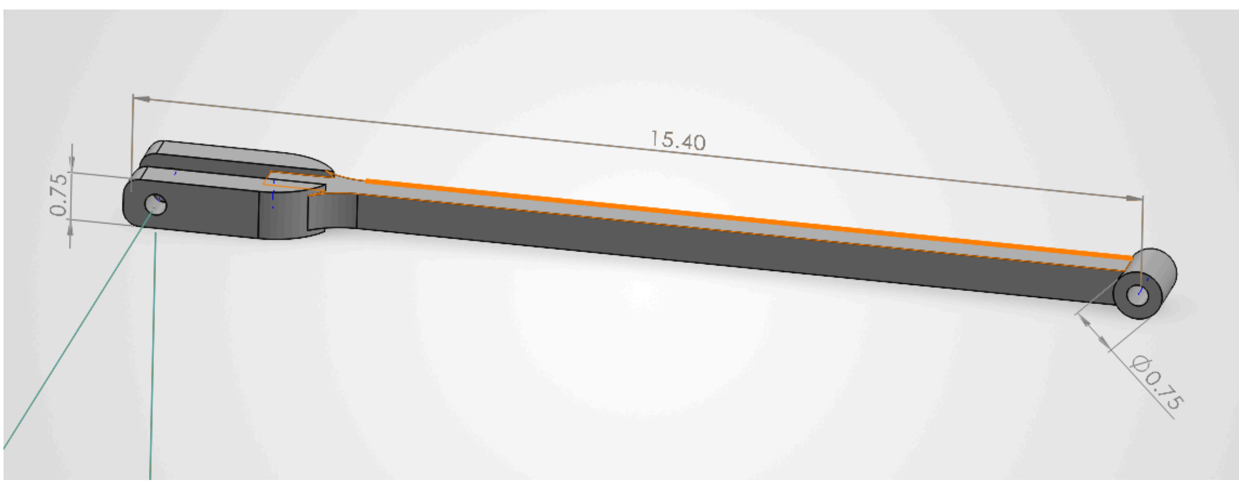
multi truss member joint with open end. Prototype 1



Prototype 2



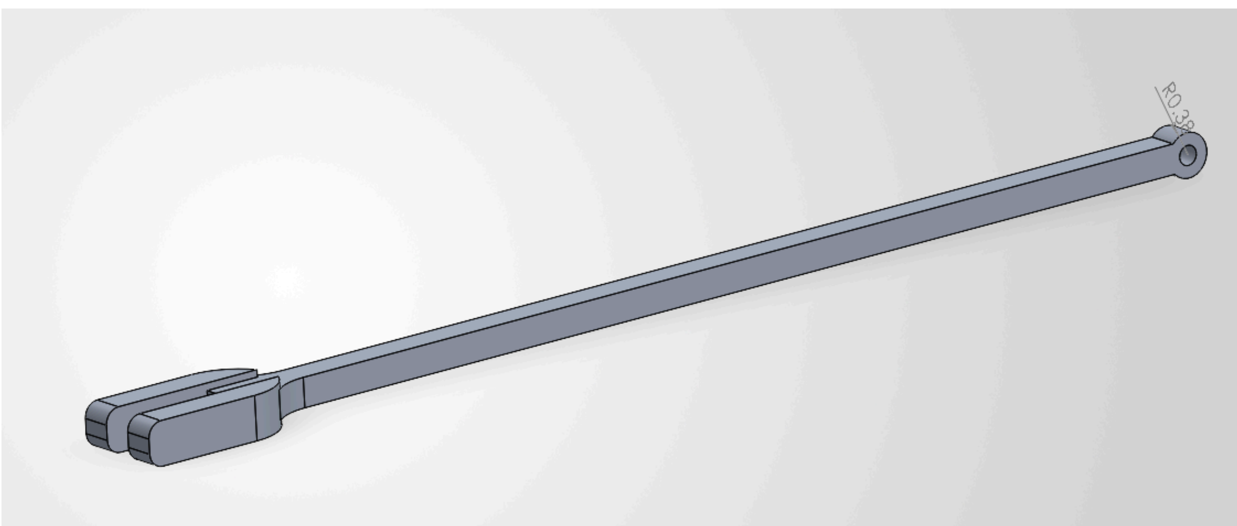
Redimensioned to match truss member length and pin diameter



Final length for 0.4m member = 15.75 inches
15.40 + Radius of closed joint end (0.375in)

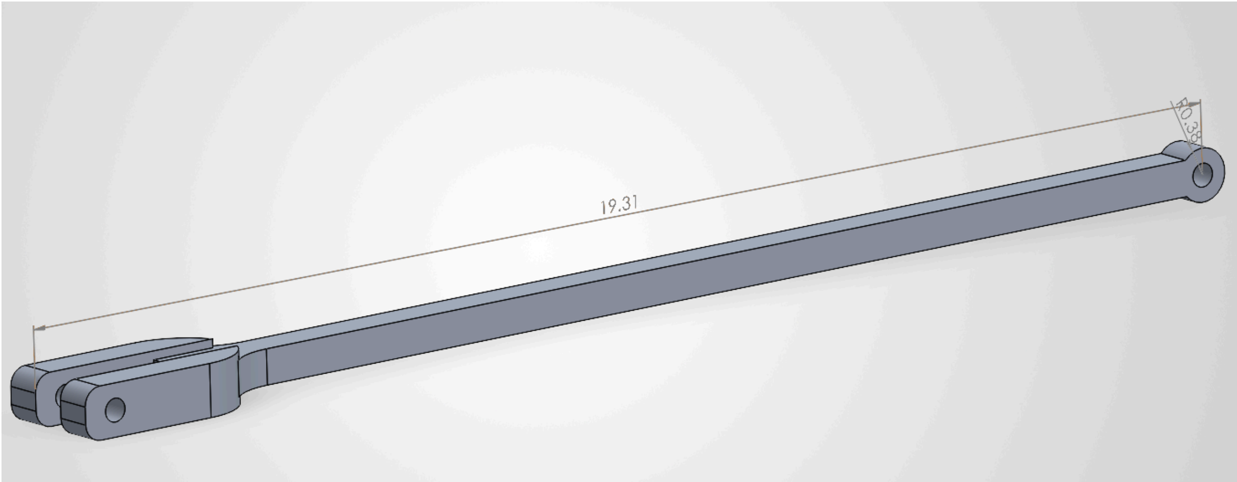
Assembly, how members are connected

Truss assembly prototype 1: need to make new members with according lengths so that top and bottom chords are parallel.

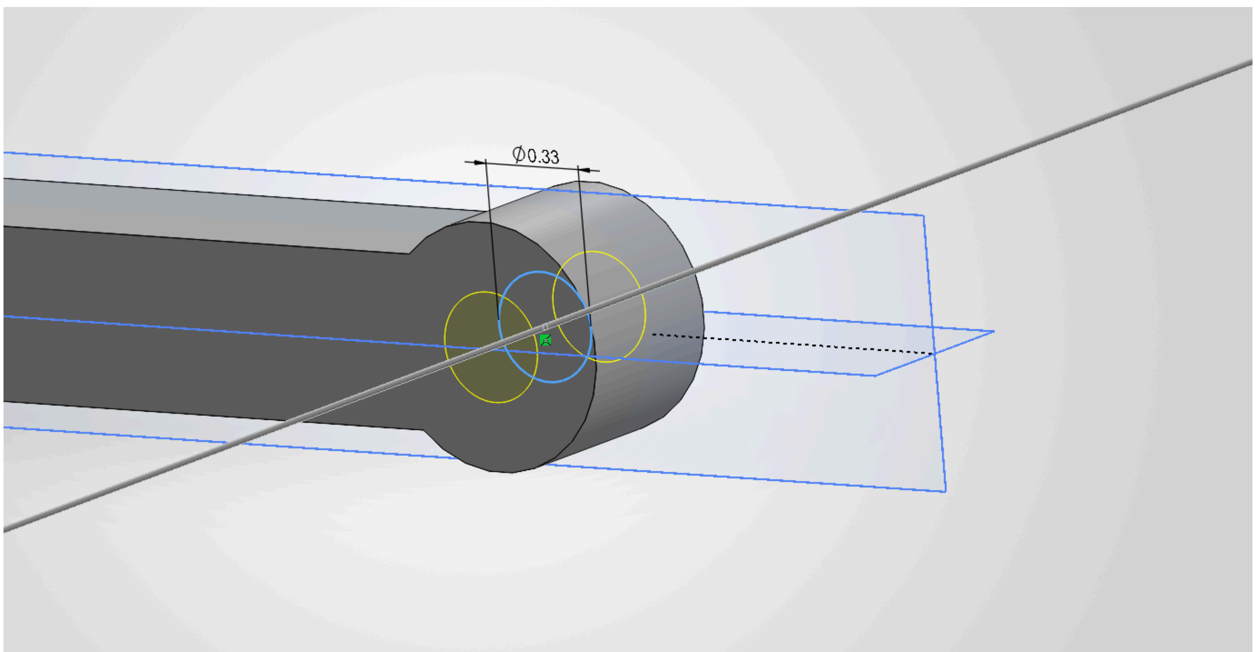


Using same features and concepts to create .5m member for my inclined parts (AD, BC)
Fillets for nicer and not so rigid look, possibly to symbolize welded members.

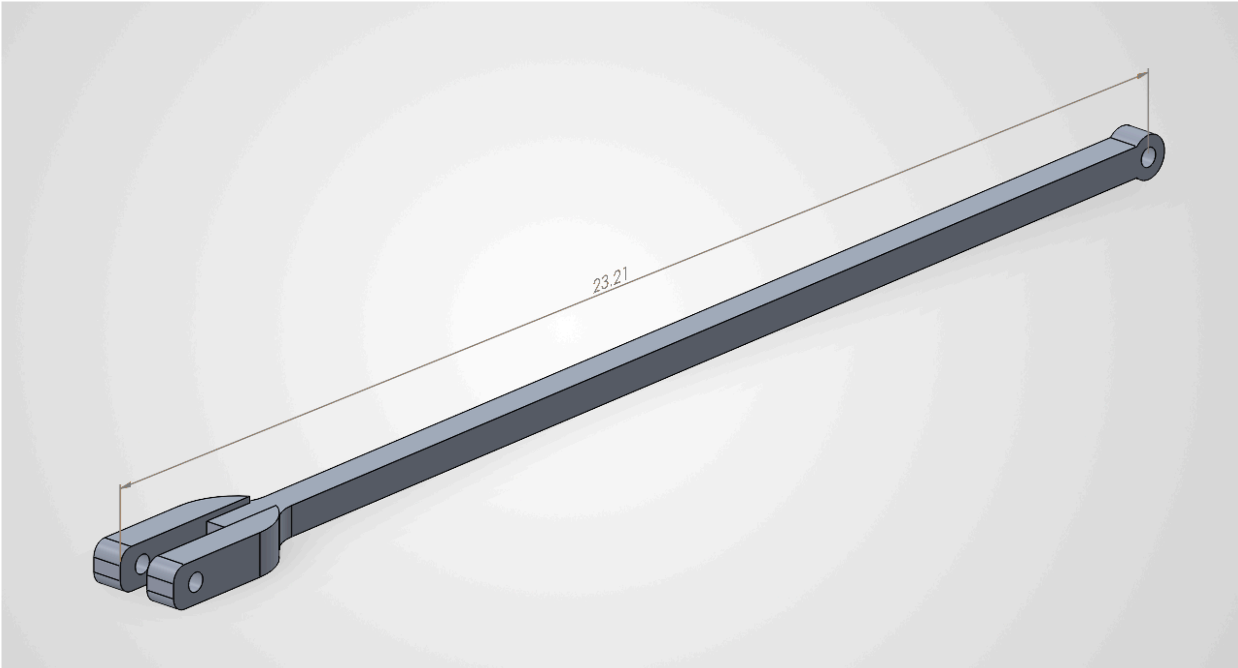
Sometimes, I had to create my own centerlines and boundaries lines because solidworks would not automatically lock into the middle of the part where I was looking to set my circle cut through or forks.



Adjusted for proper length, finished part (0.5m member)



Cutting the holes with cut function



0.6m truss member, same size as the double closed end. Can substitute for both.

Right side of truss, assembly with forked open end members

Final design of truss assembly without joint pins with open end fork design members. Originally intended to be parallel, but upon seeing the slight arc shape at the top chord I came to an important conclusion. This arc counteracts the downward P forces and weight of members on the lower chord. Therefore created a more stable structure.

with joints

Mass properties of asmb2 w-o joints A2 Sodesign
Configuration: Default
Coordinate system: -- default --

Mass = 14.43 pounds

Volume = 50.56 cubic inches

Surface area = 377.61 square inches

Center of mass: (inches)
X = 14.13
Y = 28.14
Z = 50.65

Principal axes of inertia and principal moments of inertia: (pounds * square inches)
Taken at the center of mass.

Ix = (1.00, 0.09, 0.00)	Px = 353.75
Iy = (-0.09, 1.00, 0.00)	Py = 2067.30
Iz = (0.00, 0.00, 1.00)	Pz = 2417.75

Moments of inertia: (pounds * square inches)
Taken at the center of mass and aligned with the output coordinate system. (Using positive tensor notation.)

Lxx = 368.46	Lxy = 158.09	Lxz = 0.00
Lyx = 158.09	Lyy = 2052.59	Lyz = -0.05
Lzx = 0.00	Lzy = -0.05	Lzz = 2417.75

Moments of inertia: (pounds * square inches)
Taken at the output coordinate system. (Using positive tensor notation.)

Ixx = 48811.08	Ixy = 5895.55	Ixz = 10325.54
Iyx = 5895.55	Iyy = 41947.28	Iyz = 20567.20
Izx = 10325.54	Izy = 20567.20	Izz = 16726.52

Mass properties for truss with joints.